MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

TEST POINT, DIGITAL, KEY CODE CMH

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

This specification inactive for design after date of this specification

1 SCOPE

1 1 Scope This document establishes the requirements and test procedures for the acquisition of an electronic test point module

1 2 Classification This specification covers one class of modules

Class I as specified in MIL-STD-1389

2 APPLICABLE DOCUMENTS

2 1 Government specifications and standards Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein

SPECIFICATION

MILITARY

MIL-C-28754 - Connectors, Electrical, Modular, and Component Parts, General Specification for
MIL-M-28787 - Modules, Standard Electronic, General Specification for

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer )

2.2 Order of precedence In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence

3 REQUIREMENTS

3 1 General requirements The module shall comply with the requirements of MIL-M-28787 except as modified herein

3 2 Correlation requirements Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787

3 3 Failure rate requirements All modules acquired to this specification shall have a predicted failure rate no greater than 0 456 failures per million operating hours when predicted in accordance with MIL-STD-1389

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Naval Electronic Systems Command, ATTN ELEX 8111, Department of the Navy, Washington, DC 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter



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3 4 Performance requirements The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein

3.5.1 Key code. The module key code shall be CMH

3 5.2 Size The module size shall be 1A

3 5.3 Weight The module shall have a maximum weight of 90 7 grams

3.5.4 Contact pins The module shall have 40 contact pins positioned P1A1 through P1A40

3.5.5 Depth The module shall have no additional depth requirements

3 5 6 Holddown holes The module shall have no additional requirements for location or quantity of holddown holes.

3.5 7 Marking Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/124-1, except information normally located on the fin shall be located on the body of the module (see figure 1)

3.5 8 Female connector requirement A female connector J1A1 through J1A40 as delineated in figure 2 and equivalent to male connector P1A in electrical contact characteristics shall be provided as part of this module A grounding bushing shall be part of this connector only if needed to provide electrical contact to exposed metal surfaces The connector shall be capable of mating with a forty-pin connector identical to P1A

3 6 Circuit pin configuration The functional diagram and pin assignments shall conform to figure 3.

3 7 Acceptance test All production modules shall be tested as specified in note 1 of table I

3.8 Guide pin receptacle breakout Guide pin receptacle shall be capable of withstanding a resultant breakout force with 4.5 pounds applied, when tested in accordance with 4 4 3

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25 C end-of-life limits of table I (see 6 2)

4 QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance The quality assurance provisions of MIL-M-28787 shall apply

4.2 Acceptance tests Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787

4.3 Test equipment Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665 Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified

4.4 Test procedures The module shall be tested as specified in table I

4.4 1 Alternate test methods Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirement of MIL-M-28787

4 4 2 Order of testing Test conditions A and B shall be performed in any order



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4 4 3 Guide pin receptacle breakout test The guide pin receptacle breakout test (Q test) shall be conducted as a part of the mechanical subgroup qualification test The test shall be conducted in such a manner that when the module is clamped in a test jig it will offer no reinforcement to the guide pin receptacle in any way Insert a 0 088 ±0 001 inch (2 235 ±0 025 mm) diameter pin and a 0 138 ±0 001 inch (3 505 ±0 025 mm) diameter pin into their respective guide pin receptacle, making certain the pin seats firmly at the bottom of the receptacle, and apply a force of 4 5 pounds 1 inch from the interface surface, in both directions, on each pin along both planes

a A plane parallel to the connector longitudinal axis and perpendicular to the pin axis
b A plane perpendicular to the connector longitudinal axis and perpendicular to the pin axis

If there is evidence of cracking or degradation, the module shall be rejected for failing this test

4 4 4 Engaging and separating forces test The engaging and separating forces tests (Q test) shall be conducted as a part of the mechanical subgroup qualification test With the test sample secured to the bed of a testing machine, determine the maximum engaging and maximum separating forces The maximum engaging force shall be determined using the maximum thickness test blade on figure 6 of MIL-C-28754 Each contact shall meet the requirements of MIL-C-28754

4 5 Qualification testing Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions

4 5 1 Initial electrical tests Test conditions A and B in table I shall be performed in the order specified in 4 4 2 using the initial limits at 25° ±5°C ambient

4 5 2 Operating temperature test Not required

4 5 3 Vibration tests The vibration tests shall be conducted in accordance with MIL-M-28787

4 5 4 Life test Not required

4 5 5 Mechanical tests In addition to the applicable mechanical tests of MIL-M-28787, the guide pin receptacle breakout test specified in 4 4.3 and the engaging and separating forces tests specified in 4 4 4 shall be performed

5 PACKAGING

5 1 Packaging requirements The requirements for packaging shall be in accordance with MIL-M-28787

6 NOTES

6 1 Definitions Definitions are as specified in MIL-STD-1331

6 2 End-of-life When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.9)

6 3 Cross reference of classification Military Specification MIL-M-28787/124 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No 10001-2658737E dated 1 June 1970 Part number M28787/124-1 supersedes part numbers 2658737 and M28787/124 These parts are interchangeable

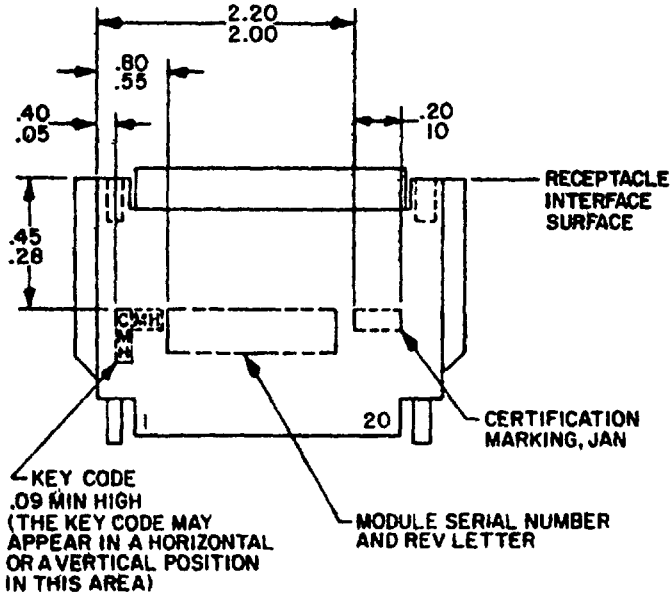
6 4 Module description The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I

Maximum power dissipation

| | |
|---|---|
| Connector pin | 5 mW |
| Module | 100 mW |
| Input current | 1 A maximum |



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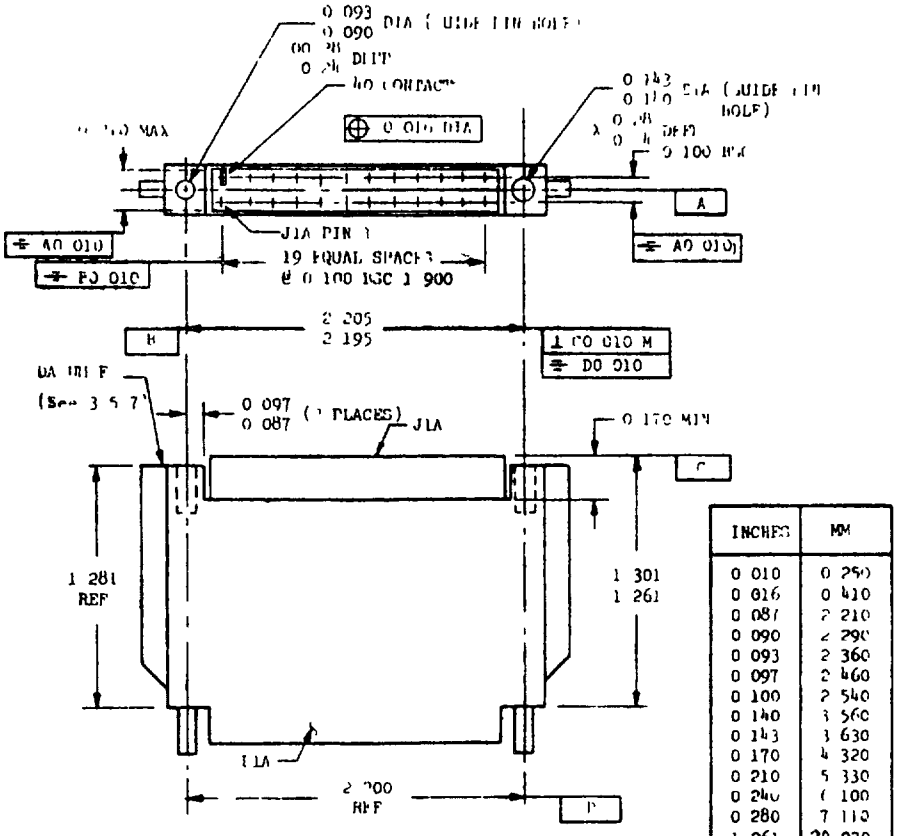



| INCHES | MM |
|---|---|
| .05 | 1.3 |
| .09 | 2 3 |
| .10 | 2.5 |
| .20 | 5.1 |
| .28 | 7 1 |
| .40 | 10.2 |
| .45 | 11.4 |
| .55 | 14.0 |
| .80 | 20.3 |
| 2.00 | 50.8 |
| 2.20 | 55.9 |

FIGURE 1. Module supplement marking information.



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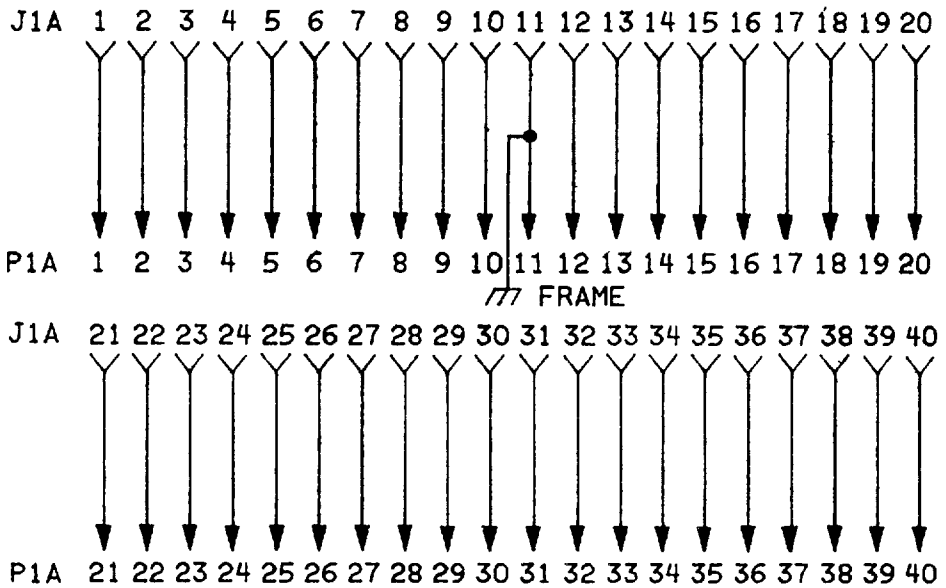



| INCHES | MM |
|---|---|
| 0 010 | 0 250 |
| 0 016 | 0 410 |
| 0 087 | 2 210 |
| 0 090 | 2 290 |
| 0 093 | 2 360 |
| 0 097 | 2 460 |
| 0 100 | 2 540 |
| 0 140 | 3 560 |
| 0 143 | 3 630 |
| 0 170 | 4 320 |
| 0 210 | 5 330 |
| 0 240 | 6 100 |
| 0 280 | 7 110 |
| 1 261 | 32 030 |
| 1 281 | 32 540 |
| 1 301 | 33 050 |
| 1 900 | 48 260 |
| 2 195 | 55 750 |
| 2 205 | 56 000 |

FIGURE 2 Female connector



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NOTES
1 J1A1-J1A40 = Female connector
2 P1A1-P1A40 = Male connector

FIGURE 3 CMH module function diagram, test point



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TABLE I Module electrical requirements

| Test condition | Parameter | Ref test table | Test Type 1/ | Acceptance requirements 25° ±5°C Ambient temperature | | | | Operating temperature 0° to 60°C | | | | Units |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| A1-A6 | Integrity | II | X | 10 | - | 10 | - | 10 | - | 10 | - | MΩ |
| B1-B41 | Continuity | III | X | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | Ω |

1/ X parameters shall be 100 percent tested during module acceptance tests



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6.5 Module function. The module consists of 39 non-isolated test points plus a chassis ground. A 40 pin female connector is attached to the top of the module in lieu of the normal fin configuration. This allows test probe monitoring of up to 39 signal points This module cannot be used as a module extender without special adapters.

6.6 Changes from previous issue Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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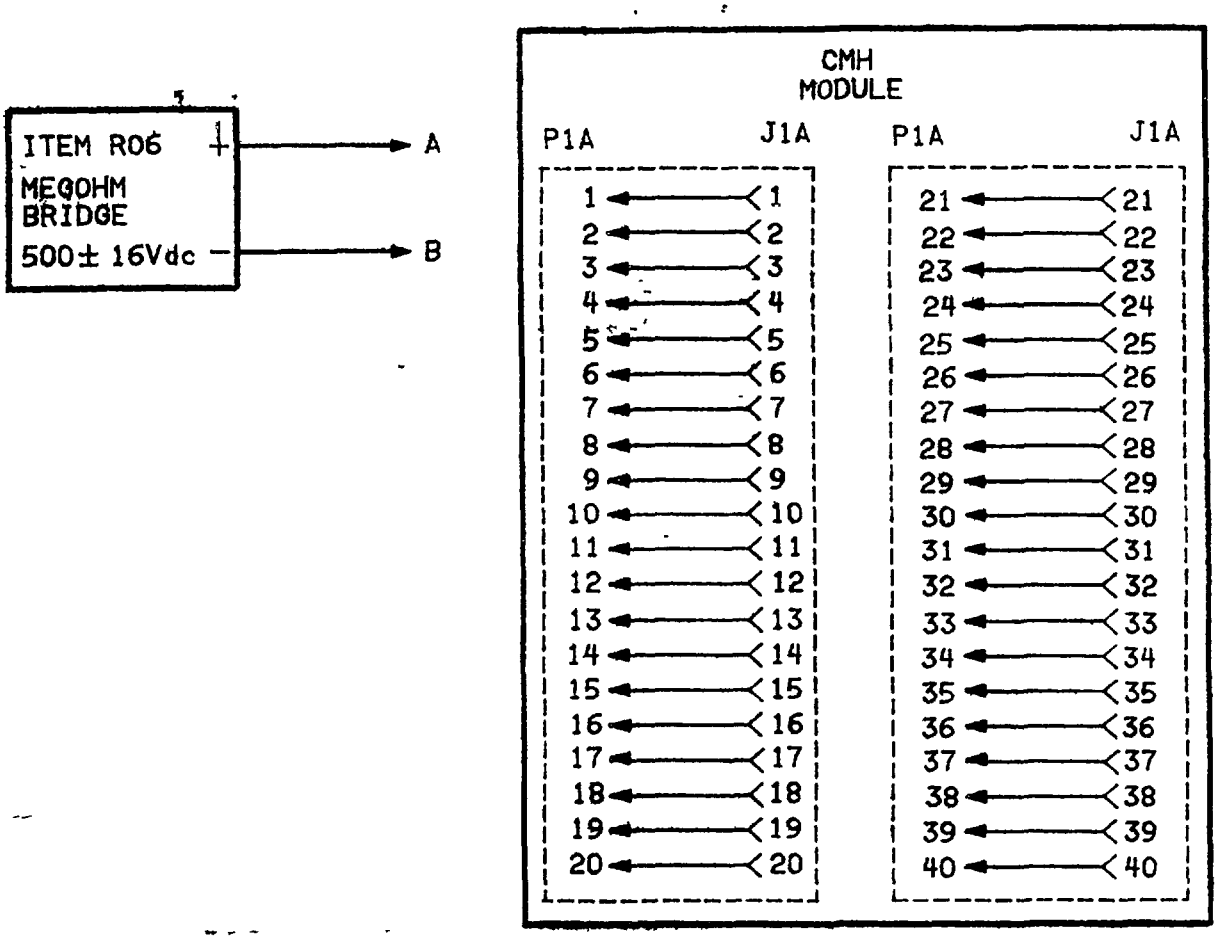
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TABLE II Integrity test.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C Ambient | | Units | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to P1A pins | B to P1A pins | Min | Max | | |
| A1 | Integrity | 1,3,5,7,9,11, 13,15,17,19, 21,23,25,27, 29,31,33,35, 37,39 | 2,4,6,8,10,12, 14,16,18,20, 22,24,26,28, 30,32,34,36, 38,40 | 10 | ~ | MΩ | 4 |
| A2 | | 3,5,8,10,13, 15,18,20,23, 28,30,33,35, 38,40 | 1,2,4,6,7,9, 11,12,14,16, 17,19,21,22, 24,25,26,27, 29,31,32,34, 36,37,39 | | | | |
| A3 | | 1,2,3,6,7,8, 11,12,13,16, 17,18,21,22, 23,26,27,28, 31,32,33,36, 37,38 | 4,5,9,10,14, 15,19,20,24, 25,29,30,34, 35,39,40 | | | | |
| A4 | | 1,2,3,4,5,11, 12,13,14,15, 21,22,23,24, 25,31,32,33, 34,35 | 6,7,8,9,10,16, 17,18,19,20, 26,27,28,29, 30,36,37,38, 39,40 | | | | |
| A5 | | 1,2,3,4,5,6, 7,8,9,10,21, 22,23,24,25, 26,27,28,29, 30 | 11,12,13,14, 15,16,17,18, 19,20,31,32, 33,34,35,36, 37,38,39,40 | | | | |
| A6 | | 1,2,3,4,5,6, 7,8,9,10,11, 12,13,14,15, 16,17,18,19, 20 | 21,22,23,24, 25,26,27,28, 29,30,31,32, 33,34,35,36, 37,38,39,40 | | | | |



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WARNING

When using the megohm bridge, high voltage may be present at any of the binding posts Always set the function switch of the megohm bridge to discharge before connecting or disconnecting the test leads

FIGURE 4. Integrity test circuit for table II



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TABLE III Continuity test

| Test condition | Parameter | Connect | | | Initial limits 25° -5°C Ambient | | Units | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to Pin | B to Pin | C to Pin | Min | Max | | |
| B1 | Continuity | P1A11 | J1A11 | Open | 0 | 1 | Ω | 5 |
| B2 | | P1A11 | Frame | Open | | | | |
| B3 | | P1A1 | C | B | | | | |
| B4 | | P1A2 | | | | | | |
| B5 | | P1A3 | | | | | | |
| B6 | | P1A4 | | | | | | |
| B7 | | P1A5 | | | | | | |
| B8 | | P1A6 | | | | | | |
| B9 | | P1A7 | | | | | | |
| B10 | | P1A8 | | | | | | |
| B11 | | P1A9 | | | | | | |
| B12 | | P1A10 | | | | | | |
| B13 | | P1A12 | | | | | | |
| B14 | | P1A13 | | | | | | |
| B15 | | P1A14 | | | | | | |
| B16 | | P1A15 | | | | | | |
| B17 | | P1A16 | | | | | | |
| B18 | | P1A17 | | | | | | |
| B19 | | P1A18 | | | | | | |
| B20 | | P1A19 | | | | | | |
| B21 | | P1A20 | | | | | | |
| B22 | | P1A21 | | | | | | |



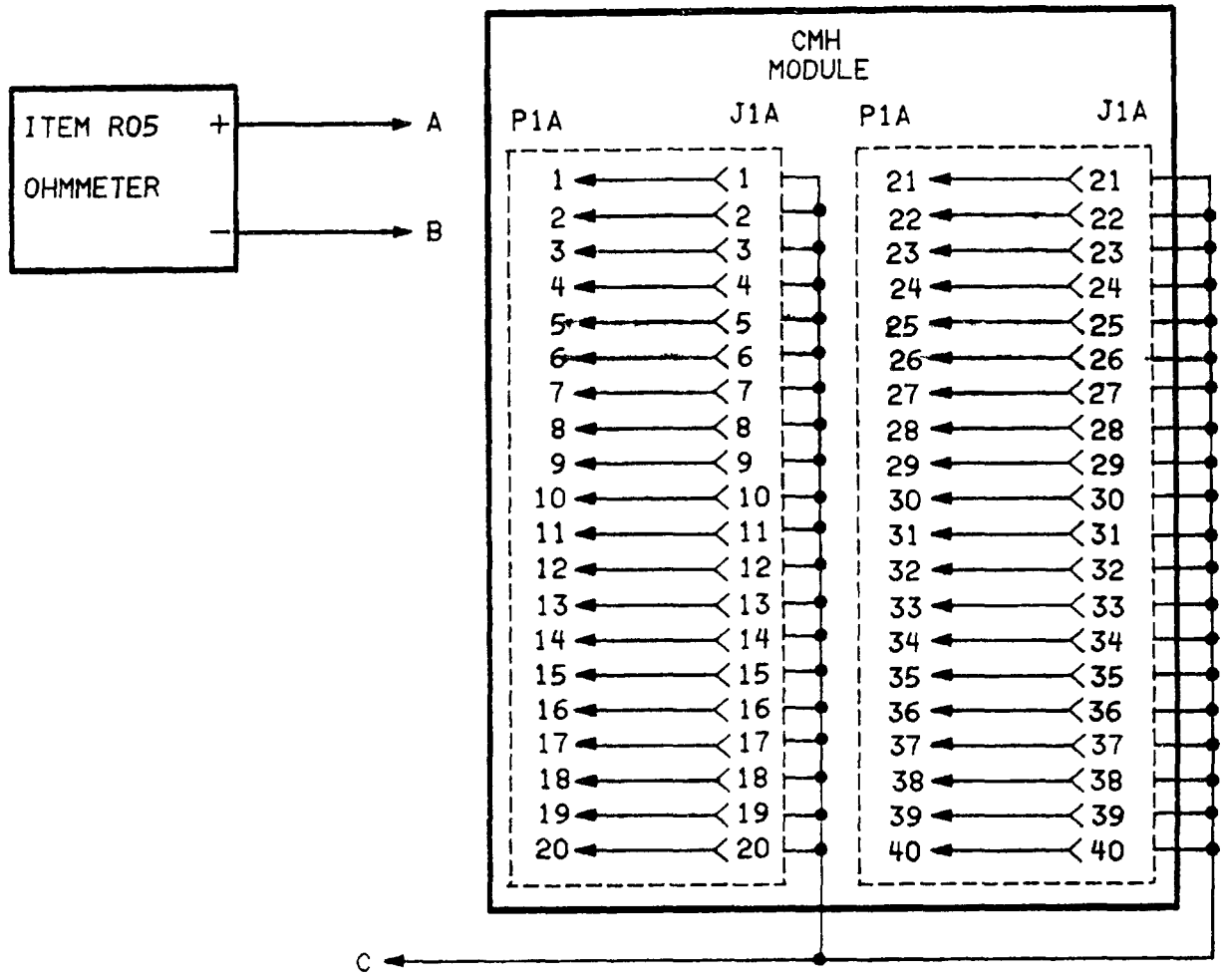
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TABLE III. Continuity test - Continued

| Test condition | Parameter | Connect | | | Initial limits 25° ±5°C Ambient | | Units | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A to Pin | B to Pin | C to pin | Min | Max | | |
| B23 | Continuity | P1A22 | C | B | 0 | 1 | Ω | 5 |
| B24 | | P1A23 | | | | | | |
| B25 | | P1A24 | | | | | | |
| B26 | | P1A25 | | | | | | |
| B27 | | P1A26 | | | | | | |
| B28 | | P1A27 | | | | | | |
| B29 | | P1A28 | | | | | | |
| B30 | | P1A29 | | | | | | |
| B31 | | P1A30 | | | | | | |
| B32 | | P1A31 | | | | | | |
| B33 | | P1A32 | | | | | | |
| B34 | | P1A33 | | | | | | |
| B35 | | P1A34 | | | | | | |
| B36 | | P1A35 | | | | | | |
| B37 | | P1A36 | | | | | | |
| B38 | | P1A37 | | | | | | |
| B39 | | P1A38 | | | | | | |
| B40 | | P1A39 | | | | | | |
| B41 | | P1A40 | | | | | | |



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NOTE To measure frame continuity, penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal

FIGURE 5 Continuity test circuit for table III



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